APMC Mandi ERP & AI Grain Assaying System

Standard Operating Procedure (SOP) & Operator Manual for Quality Inspectors, Weighbridge Operators, Branch Cashiers, Munims, and Central HQ Accountants.

> 🏛️ **APMC Mandi Deregulation & Statutory Guarantees**
> **Zero Mandi Cess (0.0%):** Hardcoded across all inward purchase bills, trader sales orders, and accounting ledgers in compliance with modern APMC market deregulation.
> **Income Tax Section 269ST:** Automated daily cash limit of ₹2,00,000 per farmer. Transactions meeting or exceeding this threshold are locked to RTGS/NEFT with mandatory Bank UTR recording, protecting trading firms from Section 271DA 100% penalties.

1. System Architecture & Role Topology

Rajeshri Shahu Mandi ERP operates on a 3-tier multi-branch topology connecting 3 active APMC market yards (**Chikhali**, **Varvand**, and **Dhad**) to a centralized HQ Audit Desk.

User Role	Assigned System Surface	Core Responsibilities
Quality Inspector	Handheld Captain Tablet (`/app`)	Sample photography, ArUco tray homography, 7-segment moisture OCR, and thermal slip printing.
Branch Cashier	Inward Weighbridge Desk (`/branch/dashboard`)	Dharamkanta weighment, QC ticket linking, Section 269ST cash compliance, and farmer settlement.
Branch Munim	Stock Register & Outward Desk	Commodity inventory tracking, vehicle dispatch, trader outward invoicing, and payment receipts.
HQ Accountant	Central Accountant Desk (`/accountant/dashboard`)	Multi-branch ledger consolidation, RTGS payout clearance, and inspector yard assignments.
Super Admin	Central Cloud Governance (`/admin/dashboard`)	Hardware-bound RSA-PSS 2048 licensing, daily mandi rate corridor discovery, and security audits.

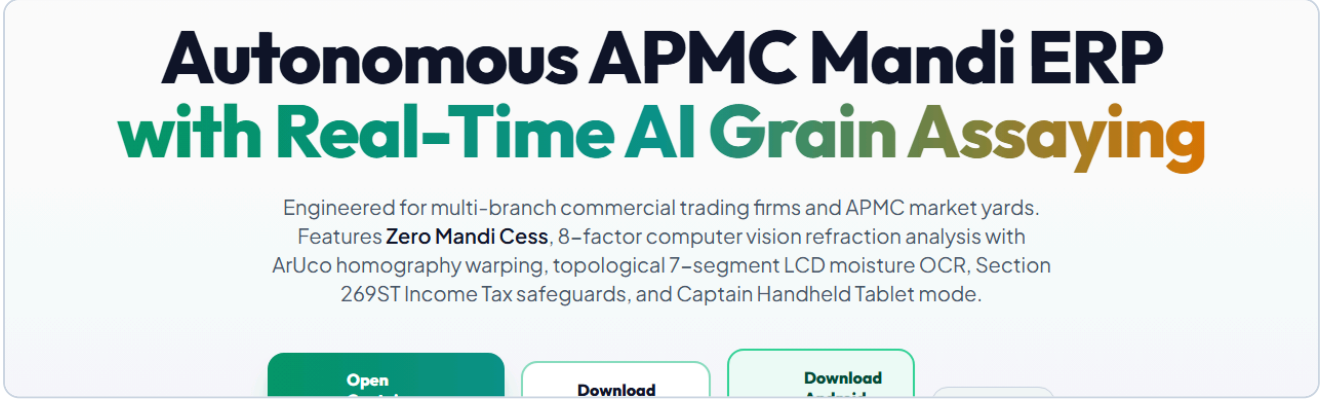


Figure 1.1: Live Mandi ERP Production Tracker & Telemetry Portal (https://mandi-erp-tracker.vercel.app)

2. Quality Inspector SOP (Captain Tablet App)

The Inspector Assaying Terminal is designed as a standalone handheld tablet and phone application, functioning like a restaurant captain order device for fast, single-purpose yard operation.

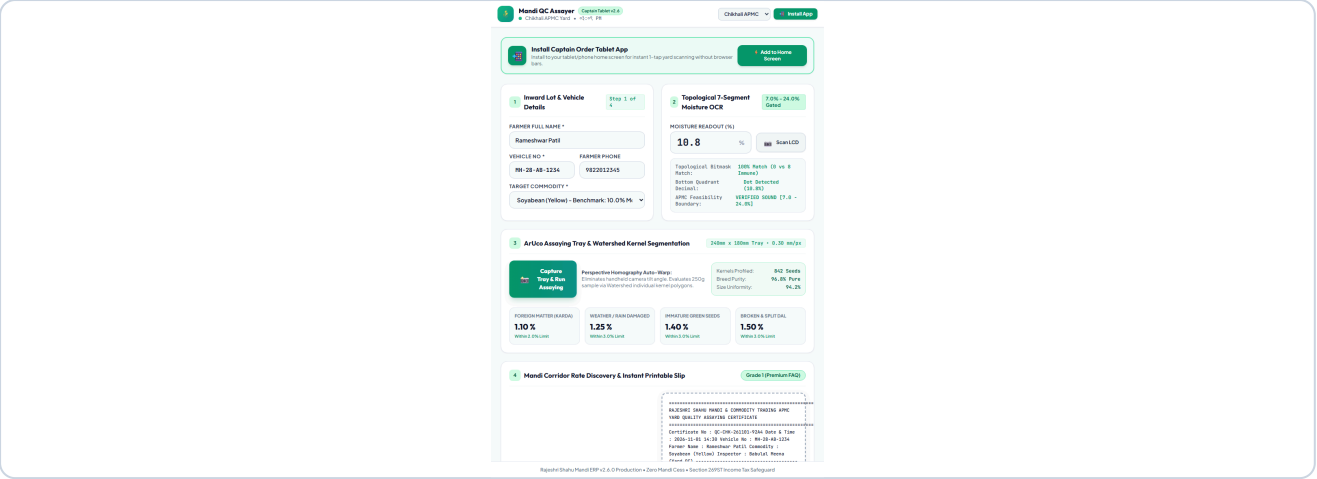


Figure 2.1: Handheld Captain Tablet Terminal UI (https://mandi-erp-tracker.vercel.app/app)

Step-by-Step Assaying Procedure:

1. **Inward Farmer & Vehicle Identification:** Enter the Farmer Full Name (e.g. `Rameshwar Patil`), Vehicle Number (e.g. `MH-28-AB-1234`), and select the target commodity (Soyabean, Wheat, Chana, Tur, Cotton, or Maize).

2. **Moisture Meter Scanning (Topological 7-Segment OCR):** Point camera at the digital display of the physical grain moisture tester (e.g., Indosaw, Kett, or Seedburo). Tap **"Scan LCD"**. The engine evaluates 7 physical bar bitmasks and searches the bottom quadrant for the decimal point.

   > **Physical Feasibility Gating:** Indian mandi moisture must strictly be between **7.0% and 24.0%**. Any reading outside this range triggers an automated re-scan prompt.

3. **Assaying Tray Capture (ArUco Homography & Watershed):** Pour the 250g sample into the standardized 240mm × 180mm matte-grey assaying tray. Tap **"Capture Tray & Run Assaying"**. The engine automatically flattens camera tilt via the 4 ArUco corner fiducials and segments each touching grain into isolated Watershed polygon contours, computing:
   - Foreign Matter / Karda % (Soil, stones, sand particles)
   - Rain & Weather Damaged % (Fungal necrosis & mold stains)
   - Immature Green Seeds % (Chlorophyll excess)
   - Broken & Split Dal % (Fracture edge contrast & aspect ratio)
   - Size Uniformity Index & Varietal Breed Purity score

4. **Corridor Rate Discovery & Printable Slip Issuance:** The system compares refractions against today's live Mandi Buying Rate Corridor and assigns Grade 1 (Premium FAQ), Grade 2 (Standard FAQ), or Grade 3 (Non-FAQ). Tap **"Print 80mm Slip"** for the thermal receipt, and tap **"Sync to Weighbridge"** to transmit data to the cashier desk.

3. Weighbridge Operator & Cashier Desk SOP

The Branch Cashier operates the primary inward desk (`/branch/dashboard`), executing vehicle weighments and farmer payment settlements.

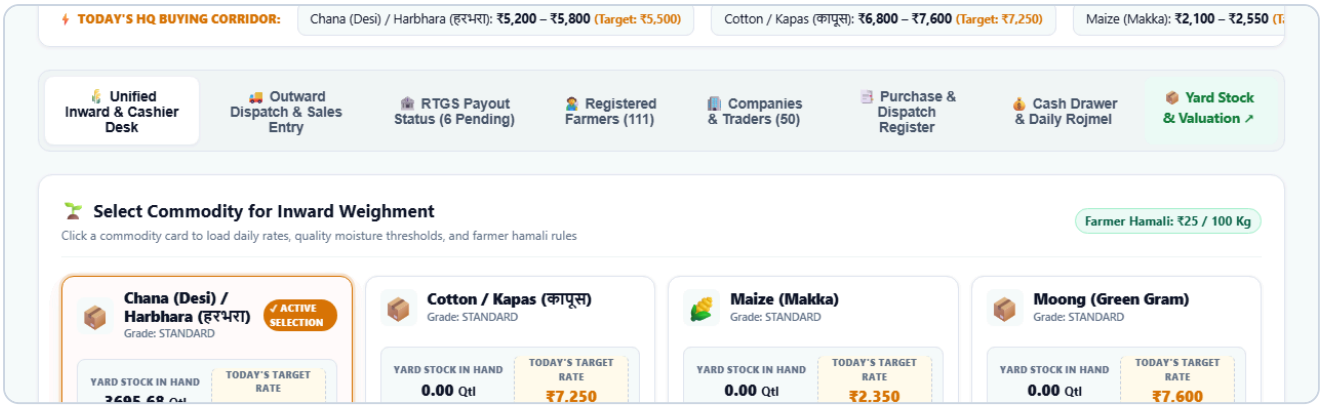


Figure 3.1: Branch Cashier Weighbridge Inward Desk & Payout Calculator

Step-by-Step Cashier Settlement Workflow:

1. **Gross Vehicle Weighment:** Loaded vehicle drives onto Dharamkanta platform. Enter Gross Weight (e.g. `3,850 Kg`) or sync directly via serial RS-232 weighbridge indicator.

2. **Link Digital QC Ticket:** Select the farmer's pending inspection ticket from the dropdown or click **"Verify & Fill"**. Moisture %, Karda %, Breed, and Suggested Rate auto-populate instantly.

3. **Tare Weighment & Net Calculation:** After grain is unloaded into the yard stack, empty vehicle mounts the weighbridge for Tare Weight (e.g. `1,250 Kg`). System calculates Net Weight: `Net Weight = Gross (3,850) - Tare (1,250) = 2,600 Kg (26.00 Qtl)` Standard moisture and karda refractions are deducted automatically from payable weight.

4. **Section 269ST Compliance & Cashier Payout:** Select payment mode (CASH or RTGS).

   > **Automated ₹2,00,000 Cash Gate:** If today's cumulative payments to this farmer reach or exceed ₹2,00,000, CASH mode is locked. The cashier must record the Bank Name, Account Number, and 16-character RTGS/NEFT UTR Number.

   For cash payments under ₹2,00,000, specify the exact currency denomination breakdown (₹500, ₹200, ₹100 notes) for automated register reconciliation.

4. Branch Munim Manual (Stock Register & Outward Sales)

The Branch Munim maintains commodity physical stock balances and invoices outward dispatches to oil mills, dal millers, and external trading firms.

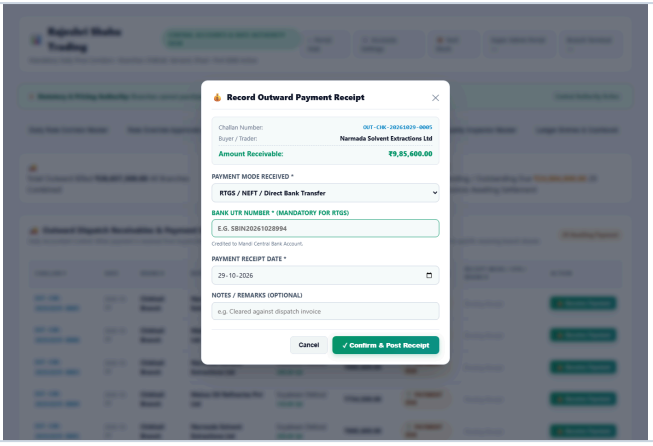


Figure 4.1: Outward Trader Invoicing & Payment Receipt Modal

Stock Book & Outward Billing SOP:

- **Automated Stock Ledgers:** Inward purchases automatically increment commodity stock registers. Unloading tare confirmations update physical godown bags in real time.
- **Outward Dispatch Invoicing:** Enter buyer trader name, GSTIN, destination, transporter details, vehicle number, and agreed rate per quintal.
- **Payment Receipt Tracking:** Munims record incoming trader payments (RTGS, Cash, Cheque) with transaction vouchers and issue printed outward dispatch passes.

5. HQ Accountant & Central Super Admin Governance

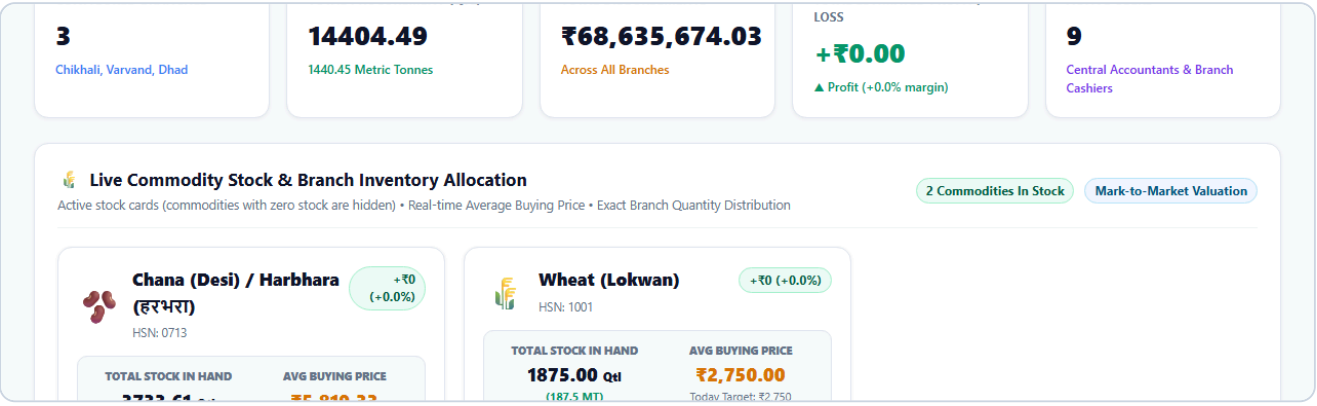


Figure 5.1: Central Super Admin Governance, Licensing & Staff Role Segregation

> 🔐 **Hardware-Bound RSA-PSS 2048 Licensing**
> The ERP license is cryptographically tied to the server CPU fingerprint and machine UUID. To activate a branch terminal, Super Admin generates a signed `.lic` key from the Central License Desk.

Staff Role Assignment SOP:

Quality inspectors are assigned to specific branches by the Accountant HQ. When Chikhali yard has 2 or 3 inspectors, each inspector is provisioned with a distinct username and password, ensuring individual audit

accountability for every issued QC certificate.

6. Technical Troubleshooting & Support

Observed Issue	Root Cause	Recommended Resolution
Moisture reading rejected	Reading is <7.0% or >24.0%	Wipe moisture sensor cell, re-insert grain cup, and tap re-scan.
Tray corners not recognized	Grain spilled over ArUco markers	Ensure 4 corner fiducial markers on matte-grey tray are unblocked.
Cash settlement disabled	Section 269ST ₹2,00,000 threshold reached	Select RTGS/NEFT mode and enter Bank UTR number. Cash is legally barred.